Sell together, fund the season, store smart

Making Kosi Agri Partners' 12 Farmer Producer Organisations pay their own way by 2030-31

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1. **Sell together**
2. **Fund the season**
3. **Store smart**

Today a KAP FPO earns 22% of what it costs to run; our plan lifts that to 92% to 98% by 2030-31 and builds a federation that pays for itself

THE PROBLEM

- A typical FPO turns over **Rs 27.5 lakh** and keeps about **Rs 1.4 lakh**, against **Rs 5.9 to 6.3 lakh** to run with a paid CEO and proper accounts.
- Its Rs 10 lakh working capital is just members' share capital (500 x Rs 2,000), against a Rs 25 lakh peak need.
- Government support to KAP, a CBBO, ends in 2027-28. Nationally only about **1 in 3** FPOs is viable.

WHAT WE RECOMMEND

1. **Sell together:** a shared sales desk sells combined lots to at least three buyers, none above half of sales.
2. **Fund the season:** claim the government matching equity grant, then a revolving fund that pays farmers within 7 days.
3. **Store smart:** lease idle cooperative (PACS) godowns; dryers for maize, solar cold rooms for vegetables.

All three run through a **federation of the FPOs** that keeps services going after grants end. US$1 million a year: about half makes the 12 FPOs self-running; the rest takes the model to **24 more Kosi FPOs**, released only if a Year 2 checkpoint is met.

92% to 98%

running cost covered by own income in Year 5 (22% today)

Rs 1.03 crore

average turnover per FPO (Rs 27.5 lakh today)

158%

of the federation's own running cost covered

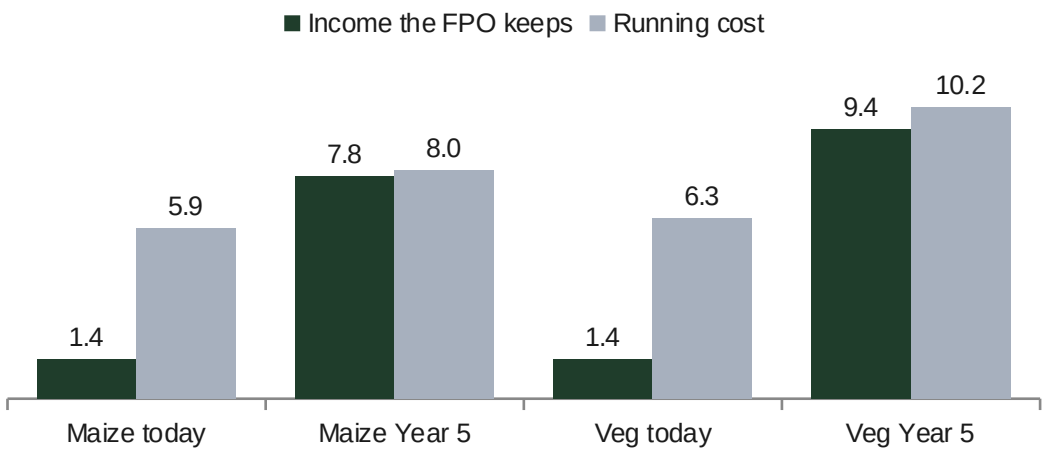
SCORECARD, END OF YEAR 5 (2030-31)

Test	Today	Target	Benchmark
Average turnover per FPO	Rs 27.5 lakh	**Rs 1.03 crore**	12.7% of scheme FPOs exceed Rs 1 crore
FPOs covering 100% of running cost, 2 years running	0 of 12	**6 of 12; all above 85%**	About 1 in 3 nationally
Members selling through the FPO	About 15%	**48%**	SRIJAN: 39% after 2 years
Women members / women directors	20% / 1 of 9	**38% / 3 of 9**	Women-led FPOs survive 4 to 13 points more often
FPOs with a credit line in their own name	0	**12**	Only 30% of FPOs have one
Federation pays its own running cost	No federation	**Yes**	Model: 158%

Rs lakh unless stated. FPO = Farmer Producer Organisation; KAP = Kosi Agri Partners; CBBO = Cluster-Based Business Organisation. Sources: case; our model; S1, S3, S4, S9, S10 (Appendix A6).

Volume is the root problem: too little produce flows through each FPO to pay its CEO, so we fix selling and seasonal finance first

INCOME KEPT VS RUNNING COST PER FPO (RS LAKH A YEAR)



WHAT THE NUMBERS SAY

- **Margins are thin** (3% on maize, 7% on vegetables), so only volume pays fixed costs. Break-even turnover is about **Rs 1.07 crore** (maize) and **Rs 1.10 crore** (vegetables), about 4 times today.
- **Cash, not demand, caps buying:** Rs 10 lakh against a Rs 25 lakh peak, with buyers paying in 30 to 60 days. Farmers sell to the trader who pays on the spot.
- **No accounts, no bank loan, no exit:** without clean books and filings no lender will fund the FPO after KAP leaves in 2027-28.
- **Infrastructure is an enabler, not a priority on its own:** it pays only where it adds sales (drying for maize, cold rooms for vegetables).

OUR THREE PRIORITIES, AND WHERE MAIZE AND VEGETABLE FPOS DIFFER

Priority	Why it matters most	Common to all 12	Maize FPOs (7)	Vegetable FPOs (5)
1. Market access and volume	Only volume covers the fixed cost of running an FPO	Written buyer agreements; honest weighing; pay farmers within 7 days	Lots of 1,000 t+ for feed mills and distilleries; moisture 14% or lower; hold some to sell later	Daily small, perishable lots; grading, crates, cold rooms; sell via VEGFED and processors
2. Seasonal finance	Cash limits how much the FPO can buy, and late pay drives farmers away	Claim matching equity grant; revolving fund; bank line	One big April to June buying season; stored maize financed by warehouse receipts	Smaller, year-round need; input credit for seed and saplings
3. Management and member participation	Independence needs accounts, compliance and members who sell through the FPO	Shared accountant via federation; CEO paid from margin by Year 5; women in jobs and on boards	Women procurement agents at collection points	Women run nursery and grading

Rs lakh a year per FPO. Running cost = CEO, assistant, office, federation fee, asset upkeep and loan repayment, with no grant. Sources: case; model sheet Unit_Econ.

Three interventions, each built to leave something behind that keeps earning after the grant stops

1 Sell together

A sales desk (KAP staff in Years 1 to 3, federation staff from Year 4) signs written buyer agreements. FPOs weigh honestly, test moisture and pay farmers within 7 days. Two paid women procurement agents per FPO, recruited from SHGs.

MAIZE

Combine 7 FPOs into lots of 1,000 t+. Feed mills, distilleries, and later sale via e-NWR; no buyer above 50%. Rs 10 lakh kit: drying yard, moisture meter, shared dryer.

VEG

Three collection points, crates, pack shed on a raised plinth, two 5 t solar cold rooms, nursery (Rs 40 lakh kit). Sell to VEGFED, processors, city traders.

Leaves behind: Buyer relationships and a desk the federation pays for.

2 Fund the season

1) Clean share registers and claim the government matching equity grant (Rs 13 lakh per FPO). 2) A revolving fund owned by the federation: Rs 2 crore plus a Rs 0.5 crore flood window, lending at 10%. 3) A Rs 0.5 crore first-loss deposit opens bank lines in each FPO's own name.

MAIZE

One big April to June buying season. Stored maize is financed by e-NWR pledge loans backed by a government guarantee.

VEG

Smaller, year-round need. Input credit for seed and saplings, repaid in produce.

Leaves behind: A fund that keeps lending, and a credit history.

3 Run it together

A federation of the FPOs gives accounts, audit, filings, the sales desk and bulk input buying for Rs 1.2 lakh a year per FPO. CEO salary support tapers 80%, 60%, 40%, 20%, then 0%.

WOMEN

150 new women members per FPO via SHGs; leadership fellowship for board seats; creches at collection points.

CLIMATE

Flood alerts; storage above flood level; nursery for fast replanting; flood loan window; help with BRFSY crop-aid claims (Bihar left PMFBY).

Leaves behind: Shared services that pay for themselves (158% cost cover).

Complements, not duplicates, government support: we claim the matching grant, lease the PACS godowns Bihar has just built, borrow under AIF and MIDH, and fund what stops when CBBO support ends in 2027-28.

AIF, MIDH, e-NWR (warehouse receipt), SHG (self-help group), BRFSY (state crop aid), PMFBY: see glossary, Appendix A6. Sources: S2, S5 to S8, S16, S17, S20.

By Year 5 a typical FPO does about 3.8 times today's business and covers 92% to 98% of a fully professional running cost

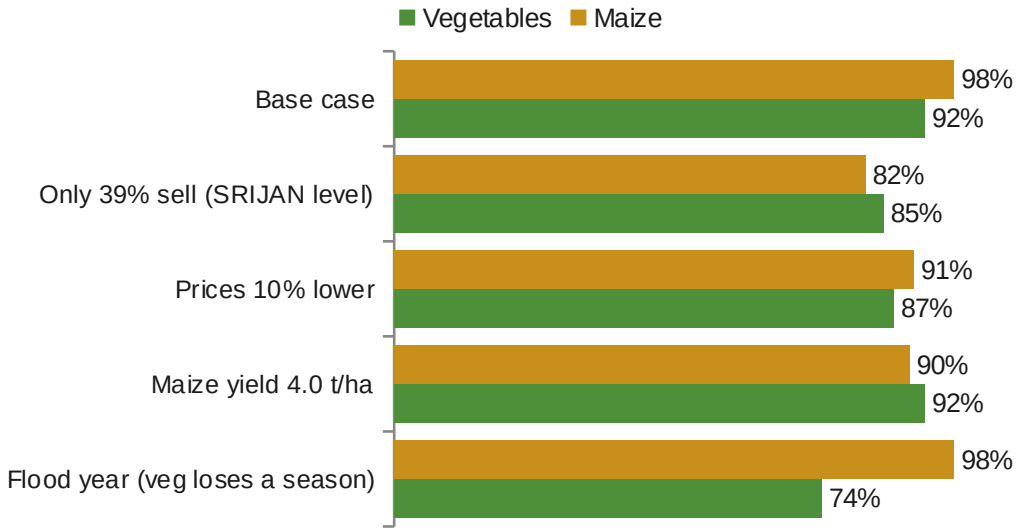
ONE FPO IN YEAR 5

Per FPO, Year 5	Maize	Vegetables
Members (including 150 new women)	650	650
Members selling through the FPO	50% (325)	45% (293)
Produce handled (tonnes)	342	559
Produce sales	77.0	78.3
Input and nursery sales	27.3	23.2
Turnover	**104.3**	**101.5**
Income kept (margins, storage gain, fees)	7.8	9.4
Running cost, no grant	8.0	10.2
Share of running cost covered	**98%**	**92%**

KEY ASSUMPTIONS

Maize: 0.4 ha a seller at 4.5 t/ha (Bihar about 4 t/ha, rabi higher); FPO handles 65% of a seller's crop; Rs 2,250 a quintal; 3% margin; 40% stored for a net 4% gain. Vegetables: 0.2 ha at 22.5 t/ha a year; FPO handles 50%; Rs 14/kg; 7% margin. Inputs 8% to 10% margin.

COST COVER UNDER STRESS

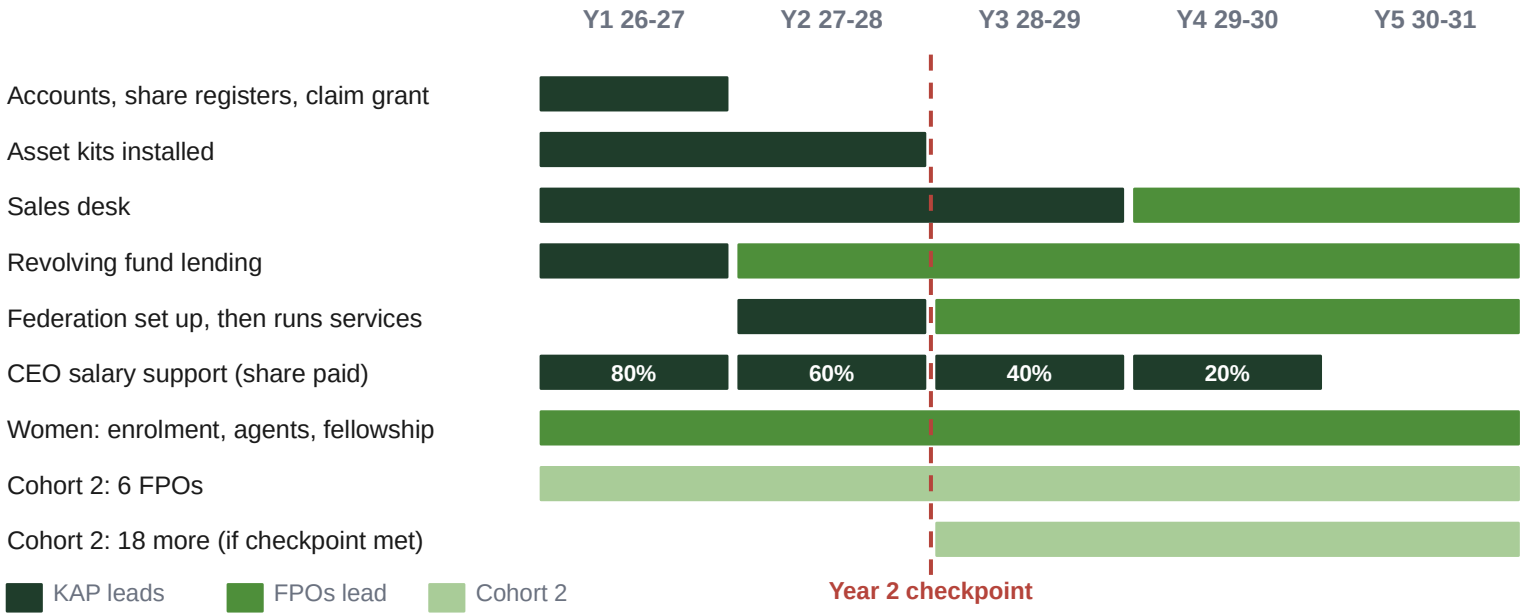


- **Targets:** average turnover Rs 1.03 crore; 6 of 12 FPOs at 100% or more for two years and all above 85%; 48% of members selling; women 38% of members.
- **Why not all 12 at 100%:** the base case sits just under break-even, so the better half must beat the average. We say so rather than overclaim.
- **What farmers get:** a fair price, honest weighing, cheaper inputs, and the option to store and sell later. Bihar's JEEViKA maize company delivered 15% to 20% higher returns; we plan below that.

Rs lakh a year per FPO unless stated. Sources: model sheets Unit_Econ and Sensitivity; assumptions and sources in Appendix A4; S9, S11, S19.

KAP builds in Years 1 and 2, hands over to the federation in Years 3 and 4, and steps back in Year 5; money continues only past a Year 2 checkpoint

PHASED ROADMAP



WHO DOES WHAT

Function	Year 1	Year 5
Finding buyers and negotiating	KAP	Federation sales desk
Accounts, audit, filings	KAP	Federation shared team
Buying from farmers and paying them	FPO with KAP	FPO CEO and women agents
Lending decisions	KAP with the Foundation	Federation credit committee with partner bank

BEFORE SUPPORT ENDS, EACH FPO MUST, FOR 2 YEARS RUNNING:

1. Cover 100% of running cost from its own income
2. Have 40%+ of members selling through it
3. Have 2+ buyers with written agreements, none above 50%
4. Hold a credit line in its own name, nothing overdue 90+ days
5. File audited accounts and returns on time
6. Have 3+ women directors and 30%+ women members
7. Have tested a flood plan in the last season

CHECKPOINT, END OF YEAR 2

Years 3 to 5 money and the 18 extra cohort 2 FPOs are released only if: **9 of 12** FPOs cover more than half their running cost; fund repayment is **95%** or more; **2+** maize buyers are in use; women are **28%+** of members. If not, the Foundation keeps the money and KAP fixes the gaps first.

Year 1 = 2026-27. Government CBBO support to KAP ends in 2027-28 (Year 2). Cohort 2 = other Kosi FPOs whose government support has ended or is ending. Sources: case; S2; our model.

About half the money makes KAP's 12 FPOs self-running; the other half takes the same model to 24 more Kosi FPOs and makes the federation self-funding

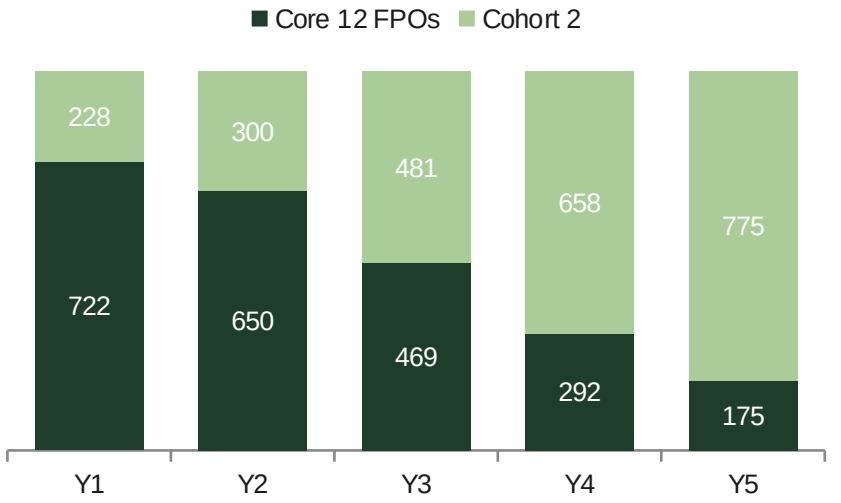
BUDGET BY YEAR (RS LAKH)

Rs lakh	Y1	Y2	Y3	Y4	Y5	5 years	Share
Sell together (sales desk, quality, women agents)	132	132	108	74	40	486	10%
Fund the season (capital that stays)	150	100	50	0	0	300	6%
Store smart (asset kits)	73	49	0	0	0	122	3%
People, women, climate	172	154	121	93	60	600	13%
KAP team, federation, evaluation	195	215	190	125	75	800	17%
Cohort 2: 24 more FPOs (18 after checkpoint)	228	300	481	658	775	2,442	51%
Total	**950**	**950**	**950**	**950**	**950**	**4,750**	**100%**
Total in US$ million	1.00	1.00	1.00	1.00	1.00	5.00	

WHO PAYS THE RUNNING COSTS AFTER THE GRANTS

Cost	Paid by	From
FPO CEO, assistant, office	FPO	Margins, storage gain, service fees (92% to 98% cover)
Accounts, audit, sales desk	Federation	Rs 1.2 lakh fee per FPO, 1% maize commission, bulk input margin, fund interest: 158% cover
Seasonal working capital	FPO and banks	Share capital plus matching grant, revolving fund, bank lines, e-NWR

CORE 12 FPOS VS COHORT 2 (RS LAKH)



Rs 38 lakh

per core FPO a year, incl. capital that stays

Rs 29 lakh

per cohort 2 FPO a year (75%: desk and fund exist)

Rs 20,300

per farmer over 5 years (23,400 members)

Why spending shifts: capital is spent early; salary and KAP support taper to zero; cohort 2 grows only after the checkpoint. The core 12 fall from 76% of the money in Year 1 to 18% in Year 5.

Rs lakh at Rs 95 per US$ (US$1 million = Rs 950 lakh a year). Sources: model sheets Budget and Federation; S2, S4, S13.

Thank you

Sell together, fund the season, store smart: 12 FPOs from 22% to 92% to 98% cost cover, a federation at 158%, and a model 24 more Kosi FPOs can use.

Unit economics: how one FPO gets from today to Year 5

Rs lakh a year	Maize today	Maize Y5	Veg today	Veg Y5
Members	500	650	500	650
Women share of members	20%	38%	20%	38%
Members selling through the FPO		325		293
Produce handled (t)		342		559
Produce sales		77.0		78.3
Input sales		27.3		18.2
Nursery sales		0.0		5.0
Turnover	**27.5**	**104.3**	**27.5**	**101.5**
Margin on produce		2.3		5.5
Net gain on stored maize		1.2		0.0
Margin on inputs		2.2		1.8
Margin on nursery		0.0		1.5
Service fees		2.1		0.6
Income the FPO keeps	**1.4**	**7.8**	**1.4**	**9.4**
CEO	3.6	3.6	3.6	3.6
Assistant	1.4	1.4	1.8	1.8
Office, travel, AGM, filing	0.9	0.9	0.9	0.9
Federation fee	0.0	1.2	0.0	1.2
Asset upkeep and insurance	0.0	0.3	0.0	1.6
Loan repayment	0.0	0.5	0.0	1.1
Running cost	**5.9**	**8.0**	**6.3**	**10.2**
Share of running cost covered	**23%**	**98%**	**22%**	**92%**
Break-even turnover at Year 5 margins		107.0		109.9

HOW TO READ IT

- Produce handled = sellers x area x yield x share sold x share the FPO handles.
- Income kept = margins on produce and inputs + storage gain + fees for drying, storage and cold rooms.
- Running cost includes the federation fee and loan repayment, so "covered" means fully independent.
- Today's cover uses the same professional cost base, to compare like with like.

Rs lakh a year. Today = case midpoint turnover; running cost today assumes a professional team with no grant. Model sheet Unit_Econ.

Working capital and asset funding: claim and lease first, then lend, then grant

WORKING CAPITAL IN YEAR 5

Rs lakh	Maize FPO	Veg FPO	All 12
Peak cash tied up in buying produce	34.7	15.7	320.9
Input credit to members	8.2	5.5	84.6
Peak need	**42.8**	**21.1**	**405.5**
Members' share capital	13.0	13.0	156.0
Government matching equity grant	13.0	13.0	156.0
Less own share of asset kit	-1.0	-4.0	-27.0
Gap to finance	**17.8**	**0.0**	**120.5**
From the federation revolving fund (60%)	10.7	0.0	72.3
From banks and NBFCs (40%)	7.1	0.0	48.2

Stored maize is financed separately by e-NWR pledge loans. NBFC = non-bank finance company. Today's gap across the 12 FPOs is Rs 180 lakh; claiming the matching grant closes most of it.

ASSET KIT PER FPO AND WHO PAYS

Rs lakh	Maize	Veg
Drying yard, moisture meter, scale	4.0	0.0
Shared mobile dryer, cleaner	5.0	0.0
PACS godown upgrade (leased)	1.0	0.0
Two 5 t solar cold rooms	0.0	18.0
Pack shed on raised plinth	0.0	8.0
Collection points and crates	0.0	8.0
Shade-net nursery	0.0	6.0
Total kit	**10.0**	**40.0**
Foundation grant	6.0	16.0
Government subsidy (MIDH 35%)	0.0	14.0
FPO own funds (10%)	1.0	4.0
AIF bank loan	3.0	6.0
Yearly loan repayment (6%, 7 years)	0.54	1.07

Assets worth 2.2 times the Foundation's grant. We lease godowns instead of building: Bihar has just built 6,158 PACS godowns (16.9 lakh tonnes).

Rs lakh. Model sheets Working_Capital and Infra. S2 matching equity grant; S5 AIF; S6 MIDH; S7 e-NWR guarantee; S16 PACS godowns.

The federation pays its own way by Year 5 (158% cost cover); FPO cost cover stays above 74% even under stress

FEDERATION IN YEAR 5 (36 MEMBER FPOS)

Rs lakh a year	Value
Federation CEO	9.0
Market desk (2 sales managers)	12.0
Accounts and compliance team	9.0
Office, software, travel, audit	5.0
Extra servicing for FPOs beyond 12	9.6
Running cost	**44.6**
Fees from member FPOs (Rs 1.2 lakh each)	43.2
1% commission on maize sold via federation	8.3
1.5% margin on bulk input buying	9.3
Net interest: Rs 159 lakh lent at 10% less 4% loss and admin	9.5
Income	**70.4**
Share of running cost covered	**158%**

FPO COST COVER UNDER STRESS, YEAR 5

Scenario	Maize	Veg
Base case	98%	92%
Only 39% of members sell (SRIJAN level)	82%	85%
Prices 10% lower	91%	87%
Maize yield 4.0 t/ha	90%	92%
Flood year: vegetables lose 1 of 1.5 seasons	98%	74%

Revolving fund: Rs 250 lakh capital (incl. Rs 50 lakh flood window); about 64% lent out on average, so the flood window stays liquid. Fund rate of 10% is below the 14% to 15% NBFCs charge FPOs.

Rs lakh a year. Model sheets Federation and Sensitivity. S4 NBFC lending rates.

Assumptions behind the numbers, with reasons and sources

Assumption	Maize	Veg	Reason and source
Members who sell produce through the FPO	50%	45%	[S9] SRIJAN raised members selling to their FPOs to 39% in 2 years; we allow 5 years plus input credit tied to sales. Vegetables lower because of daily, small lots
Crop area per selling member	0.4 ha	0.2 ha	[S11] 83% of Bihar holdings are under 1 ha. Maize is the main rabi crop in Kosi; vegetables are grown on small plots
Yield per hectare per year	4.5 t/ha	22.5 t/ha	[S11] Bihar maize about 4 t/ha, rabi higher. Vegetables: 15 t/ha a season x 1.5 seasons (our assumption)
Share of a selling member's output the FPO handles	65%	50%	Input credit repaid in produce and assured payment in 7 days pull volume; vegetables lower because of side-selling to local markets
Average farm-gate price	Rs 22,500/t	Rs 14,000/t	[S14] Maize MSP Rs 2,400/qtl but Bihar has no state maize procurement; Purnia mandi near Rs 2,100 to 2,300. Vegetables: blended Rs 14/kg (our assumption)
Input spend per member per year	Rs 12,000	Rs 8,000	Seed, fertiliser and crop protection for 0.4 ha maize or 0.2 ha vegetables (our assumption)
Margin on produce sold	3%	7%	Maize: about Rs 65/qtl spread on bulk contracts. Vegetables: grading premium and lower loss [S12: 4.9% to 11.6% vegetable loss]
Share of maize held in storage for later sale	40%	0%	[S15] Small farmers sell in distress at harvest; larger farmers hold 3 to 4 months
Net gain on stored maize after rent, interest and shrinkage	4%	0%	Assumes about 9% to 10% price rise in 4 months less about 5% holding cost; e-NWR pledge loan pays for the stock [S7]
Margin on inputs	8%	10%	Dealer margin on bulk seed and fertiliser (our assumption)
CEO salary	Rs 3.6 lakh	Rs 3.6 lakh	Rs 30,000 a month. Govt management cost norm is Rs 18 lakh over 3 years = Rs 6 lakh a year for all costs [S2]
Fee to federation for accounts, compliance and market desk	Rs 1.2 lakh	Rs 1.2 lakh	Shared chartered accountant, audit, ROC filings and sales desk; cheaper than each FPO hiring its own
Asset kit per FPO	Rs 10 lakh	Rs 40 lakh	See Infra sheet. Maize stores in leased PACS godowns, so its kit is small
Foundation grant share	60%	40%	Our choice: grant share is higher for maize because its kit earns less
Peak working capital as share of yearly produce sales	45%	20%	Maize is one big rabi buying season with 30 to 60 day buyer payment; vegetables turn over within days
Exchange rate	Rs 95		[S13] Rupee at 94.8 to 95.95 per USD in Aug to Sep 2026; 95 used

Full list with notes in model sheet Assumptions. Source codes refer to Appendix A6.

Evidence and hit rates behind the targets, and the risks we watch

WHAT SIMILAR EFFORTS ACHIEVED

Evidence	Figure
FPOs that are financially viable (ICAR-NIAP)	About 1 in 3
Scheme FPOs above Rs 1 crore turnover, FY25	1,135 of 8,961 (12.7%)
FPOs profitable only with support (NAFPO 2026)	45%
Profitable without support: with vs without a loan	61% vs 43%
Women or mixed-led FPOs: survival after support	4 to 13 points higher
SRIJAN, 12 FPOs, 2 years: turnover; members selling	+190%; 39%
JEEViKA maize company (Purnia): farmer returns	+15% to 20%
Producer companies older than 7 years, moribund	45%

RISKS: WARNING SIGNS AND RESPONSES

Risk	Warning sign	Response
Ethanol demand cut	Distillery delays lifting or pays after 30 days	Three-buyer rule; shift to feed mills; store under e-NWR
Side-selling to traders	Members selling stalls below 30%	Pay in 7 days; input credit repaid in produce; daily prices
Bad loans	Repayment below 95%	Credit committee; first-loss deposit; stop new loans to that FPO
Flood	Kosi alert or early inundation	Raised godowns; flood loan window; nursery replanting; BRFSY claims
Elite capture	Same few directors; women fellows not elected	Rotation rules; women's seats in exit tests; independent audit
KAP spread thin	Cohort 1 milestones slip	Cohort 2 expansion gated on the checkpoint

Sources: S1, S3, S4, S9, S10, S18, S19, S21, S22 (Appendix A6). S9 is self-reported; S22 is a secondary citation.

Sources and glossary

S1 ICAR-NIAP via Business Standard, Aug 2025: one third of FPOs viable
S2 Govt 10,000 FPO scheme guidelines and Rajya Sabha replies (CBBO fee, management cost, matching equity grant)
S3 Ministry of Agriculture via IANS, Jul 2025: FPO turnover bands, FY25
S4 NAFPO State of the Sector 2026 (Tribune); NABARD credit guarantee scheme; Samunnati FPO loan rates
S5 Agriculture Infrastructure Fund (myScheme, Indian Bank)
S6 MIDH post-harvest assistance (Rajya Sabha, Mar 2025)
S7 CGS-NPF e-NWR pledge finance guarantee, Dec 2024
S8 Solar cold room service, Muzaffarpur (pv magazine India)
S9 SRIJAN and Walmart Foundation results (Agrospectrum, Oct 2025)
S10 TCI Cornell and BIRD, State of FPOs in India 2026
S11 Maize supply chain in Bihar (DES); BAMETI state profile
S12 NABCONS 2022 post-harvest losses (Rajya Sabha, Feb 2025)
S13 Rupee at 94.8 to 95.95 per US$, Aug to Sep 2026
S14 Maize MSP 2025-26; Gulabbagh mandi prices
S15 Village Square, 101Reporters: distress sale of maize at harvest
S16 Bihar PACS godowns, Jun 2025
S17 Bihar exit from PMFBY; BRFSY terms
S18 Mongabay, Feb 2026: Bihar ethanol allocation cut
S19 World Bank and JEEViKA: Aaranyak maize company
S20 VEGFED and Tarkaari
S21 Sahyadri Farms FY24
S22 MCA data on moribund producer companies (secondary)

GLOSSARY

FPO Farmer Producer Organisation
KAP Kosi Agri Partners
CBBO Cluster-Based Business Organisation
PACS Primary Agricultural Credit Society
AIF Agriculture Infrastructure Fund
MIDH Mission for Integrated Development of Horticulture
e-NWR electronic Negotiable Warehouse Receipt
CGS-NPF credit guarantee for e-NWR pledge finance
NBFC non-bank finance company
SHG self-help group
JEEViKA Bihar Rural Livelihoods Project
VEGFED Bihar State Vegetable Processing and Marketing Cooperative Federation
BRFSY Bihar Rajya Fasal Sahayata Yojana (state crop aid)
PMFBY Pradhan Mantri Fasal Bima Yojana (central crop insurance)
MSP minimum support price

Benchmarks confirmed through published reports and news coverage; full links in the working document.